

**main.py** diagram.json ssd1306.py ▼

```
1  from machine import Pin, ADC, I2C
2  import ssd1306
3  import time
4
5  # ADC setup
6  current_adc = ADC(Pin(34))
7  voltage_adc = ADC(Pin(35))
8
9  current_adc.atten(ADC.ATTN_11DB)
10 voltage_adc.atten(ADC.ATTN_11DB)
11
12 # I2C OLED
13 i2c = I2C(0, scl=Pin(22), sda=Pin(21))
14 oled = ssd1306.SSD1306_I2C(128, 64, i2c)
15
16 energy = 0
17 last_time = time.ticks_ms()
18
19 def read_stable adc):
20     total = 0
21     for _ in range(10):
22         total += adc.read()
23         time.sleep_ms(2)
24     return total / 10
25
26 while True:
27     raw_current = read_stable(current_adc)
28     raw_voltage = read_stable(voltage_adc)
29
30     # Convert values
31     current = (raw_current / 4095) * 10    # 0-10A
32     voltage = (raw_voltage / 4095) * 250  # 0-250V
33
34     power = voltage * current
```